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 Oefeningenreeks 1b nr. 24.9

Bereken voor $a, b \in \mathbb{R}$:

$$\begin{aligned}
 & 1 + \frac{a-bi}{a+bi} + \frac{a+bi}{a-bi} \\
 &= 1 + \frac{a-bi}{a+bi} \cdot \frac{a-bi}{a-bi} + \frac{a+bi}{a-bi} \cdot \frac{a+bi}{a+bi} \\
 &= 1 + \frac{(a-bi)^2}{a^2+b^2} + \frac{(a+bi)^2}{a^2+b^2} \\
 &= 1 + \frac{a^2-2abi-b^2}{a^2+b^2} + \frac{a^2+2abi-b^2}{a^2+b^2} \\
 &= \frac{a^2+b^2}{a^2+b^2} + \frac{\cancel{a^2-2abi-b^2} + \cancel{a^2+2abi-b^2}}{a^2+b^2} \\
 &= \frac{3a^2-b^2}{a^2+b^2}
 \end{aligned}$$

References